

WHITE PAPER

The AI Pharma Challenge

The Data Readiness Strategy for AI at Scale

AI Data Readiness for
Global Pharmaceutical Enterprises

March 2026 · v5.0

*For CIOs, AI Executives, Line-of-Business Leaders,
and Heads of Infrastructure at Global Pharmaceutical Enterprises*

Intended Audience

CIOs, AI executives, line-of-business leaders, and heads of infrastructure at global pharmaceutical enterprises. AI infrastructure decisions now carry P&L consequences that cross every function—R&D velocity, manufacturing yield, regulatory exposure, capital allocation. This document provides the shared analytical foundation these leaders need to align on a coherent strategy.

Data valid as of Q1 2026. All CapEx, regulatory, and pricing data sourced from public filings and official guidance.

Executive Summary

Pharmaceutical has more to gain from AI than almost any other industry. The question facing global top pharma today is not whether to invest, but how to deploy AI at full enterprise scale—faster and more cost-effectively than the competition—without falling into the traps that turn ambitious programs into siloed, overpriced, underpowered installations. The traps are real and documented: 42% of enterprise AI initiatives failed to meet ROI expectations in 2025. Only 11% achieved enterprise-wide implementation. EY and MIT research found 95% of manufacturing AI pilots fail to deliver measurable business value—not because the models were wrong, but because the data foundations and infrastructure were not ready for production scale.

At the same time, the external infrastructure environment is tightening. Cloud and neocloud providers are engaged in a historical race to capacity, committing over \$610 billion in 2026 CapEx—acquiring land, energy contracts, and grid connections at a pace that has turned them into extraction businesses. GPU capacity is physically scarce, pricing is structurally inflationary, and the most cloud-dependent organizations are the most exposed to rationing, repricing, and lock-in. Solving both sides—internal data readiness and external infrastructure economics—simultaneously is the challenge that defines this moment.

Global top pharma cannot afford fragmented, reactive procurement. What is needed is a comprehensive, enterprise-wide AI infrastructure strategy that allocates workloads deliberately across cloud, colocation, and owned capacity—and that positions the organization to accelerate deployment while containing cost, managing regulatory complexity, and eliminating technical debt. This report provides the foundation for that strategy.

Reference Points

- \$610B: Combined 2026 hyperscaler CapEx. Enterprise customers without reserved capacity face structural rationing.
- 42%: Of pharmaceutical AI initiatives failed to meet ROI expectations in 2025. The gap is implementation and infrastructure, not technology.
- 95%: Of AI pilots in manufacturing fail to deliver measurable business value (EY/MIT, 2025). Data foundations are the consistent root cause.
- 68%: Of pharma tech executives say neglecting data quality and governance is the primary reason AI fails (ZS/Harris Poll, 2025).
- \$4B → \$25B: Drug discovery AI investment trajectory, 2025 to 2030 (EY). The infrastructure supporting these programs must scale with them.
- 83%: Of pharmaceutical companies lack automated controls to prevent sensitive data leakage through AI tools (Kiteworks, 2025).
- 40%: Gartner projection for AI data centers restricted by power shortages by 2027.

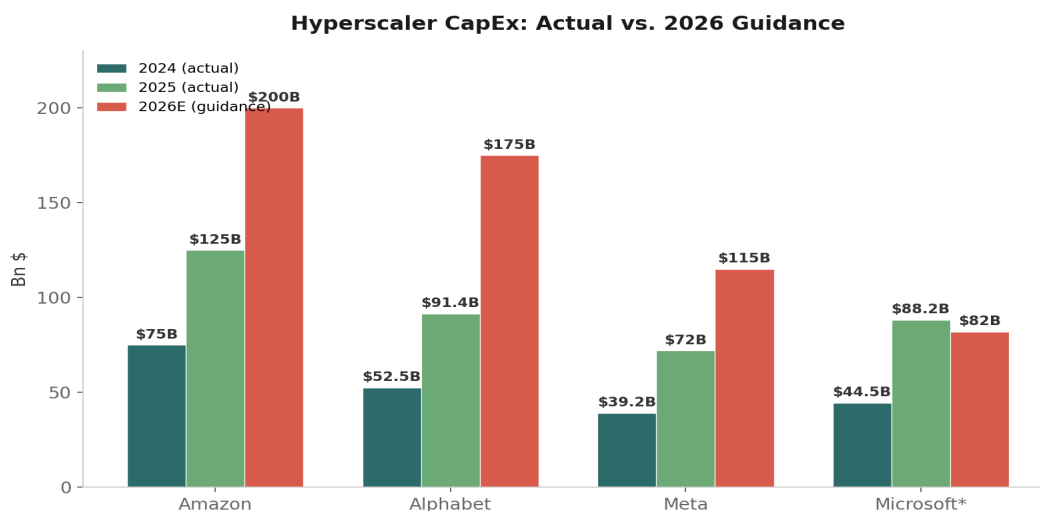
Part 1 — The Hyperscaler Landscape

1.1 CapEx at Industrial Scale

Combined hyperscaler capital expenditure has grown from approximately \$90 billion in 2020 to a projected \$610 billion in 2026—a sevenfold increase in six years. Goldman Sachs estimates current tech CapEx at 0.8% of US GDP, versus a 1.5% historical peak during the telecom bubble, implying theoretical headroom of roughly \$200 billion.

*Microsoft FY runs July–June. All figures from Q4 2025 earnings and official guidance.

Company	2024 (actual)	2025 (actual)	2026E (guidance)
Amazon	\$75.0B	\$125.0B	\$200.0B
Alphabet	\$52.5B	\$91.4B	\$175–185B
Meta	\$39.2B	\$72.0B	\$115–135B
Microsoft*	\$44.5B	\$88.2B	~\$82B (H1 annualized)



*Microsoft FY runs July–June. Sources: 10-K filings, earnings calls Q4 2025.

Figure 1 — Hyperscaler CapEx: actual vs. 2026 guidance (\$B). Sources: 10-K filings, earnings calls Q4 2025.

1.2 Cloud Providers as Extraction Businesses

These figures require interpretation. A company spending \$200 billion annually on capital assets is not operating as a technology company in any traditional sense. It is operating as an extractive industry—acquiring land, securing water rights, negotiating 15–20 year energy contracts, stockpiling transformers, and building physical infrastructure with economic lifespans measured in decades. Amazon added 3.8 GW of data center capacity in 12 months—more than any other cloud provider—and plans to double again by 2027.

The business model analogy is instructive. An extractive business controls a scarce physical resource, processes it at industrial scale, and sells access at a margin determined by supply constraints. Cloud GPU capacity operates identically: the resource (accelerator time on interconnected clusters) is physically scarce, demand exceeds supply, and pricing reflects the provider’s capital structure rather than the customer’s value equation.

For a pharmaceutical enterprise, the implications are specific. Drug discovery AI programs run for years. Clinical NLP pipelines operate continuously. Manufacturing optimization is real-time and indefinite. These are not burst workloads suited to elastic cloud consumption—they are sustained, predictable loads that generate the steady-state revenue on which extractive businesses depend. The enterprise customer is the resource being extracted.

If hyperscalers accept the logic of negative FCF and trillion-dollar debt issuance, they are implicitly betting that enterprise customers will remain dependent on cloud capacity regardless of pricing. The question for the pharmaceutical infrastructure executive is whether that dependency is a strategic choice—or an inherited condition that can be deliberately restructured.

1.3 Free Cash Flow Compression

This CapEx trajectory is compressing free cash flow across the hyperscaler sector. The pattern is most visible at Amazon, where trailing 12-month FCF declined from \$25.9 billion in Q1 2025 to approximately \$10 billion by Q4, even as quarterly CapEx rose to \$36 billion.

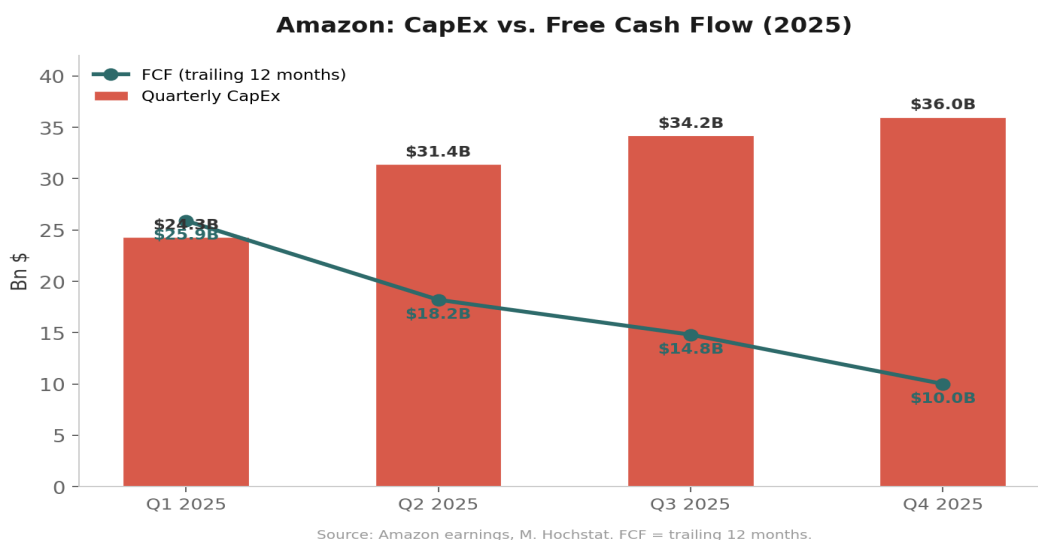


Figure 2 — Amazon quarterly CapEx vs. trailing FCF (2025). Source: Amazon earnings.

Meta projects negative FCF by 2027–2028. Collectively, hyperscalers raised \$108 billion in debt during 2025, with projections reaching \$1.5 trillion over the coming years. When the infrastructure providers funding your AI capacity are leveraging to acquire physical assets, pricing power shifts. The implications for enterprise customers—particularly those with unconsolidated subsidiary accounts—are direct and material.

The structural consequence is pricing volatility. Cloud GPU costs have increased up to 300% year-over-year in high-demand regions. Elastic scaling—once positioned as cloud’s primary advantage—now generates unpredictable monthly bills that can exceed \$10 million for a single large-scale AI program. For a pharmaceutical organization running multi-year drug discovery pipelines, this cost uncertainty is incompatible with the fixed-horizon budget cycles that govern R&D investment. The alternative is an on-premises model with fixed pricing per GPU and storage capacity consumed—where the cost of infrastructure is known at the point of commitment, not discovered at the point of invoice.

1.4 What This Means for Large Pharma

A Top-20 pharmaceutical enterprise typically holds a parent-level cloud agreement—a multi-year commitment with negotiated pricing, priority capacity, and dedicated support. This is the Tier 1 relationship.

Within the same organization, however, the landscape is fragmented. A diagnostics subsidiary, a recently acquired biotech, a regional commercial analytics team, or a manufacturing site may operate on separate

cloud accounts—often at on-demand rates or with lightly negotiated terms. These internal Tier 2 arrangements pay 60–80% more for identical compute. When Azure is capacity-constrained (confirmed H1 FY26), these accounts are the first to experience GPU queuing and service deprioritization.

The storage problem compounds this: subsidiary environments typically pair expensive GPUs with whatever storage was already in place—legacy NAS designed for file shares, not parallel AI I/O. The result is the worst of both worlds: premium cloud pricing and underutilized compute.

A subsidiary running a pharmacovigilance NLP pipeline on Azure on-demand pays ~\$3.10/hr per NC-series GPU VM. The same workload under the parent's Reserved Instance agreement costs ~\$0.93/hr. Across 50 VMs, the annual delta exceeds \$950,000—for one workload in one division. Multiply across the enterprise.

Part 2 — Cloud Reservation Structures

Understanding the contract landscape is a prerequisite for any infrastructure strategy. The table below summarizes the primary reservation mechanisms relevant to multi-geography pharmaceutical enterprises.

Provider	Mechanism	Maximum Discount	Commitment	Enterprise Pharma Context
AWS	Enterprise Discount Program	Custom (40–65%)	Multi-year spend	Parent-level; requires subsidiary consolidation
AWS	Savings Plans (Compute)	66%	1 or 3 years	Cross-region flexibility for global operations
Azure	Reserved + Hybrid Benefit	Up to 80%	3 yr + license	Maximum; requires license consolidation across entities
Azure	Enterprise Agreement	Custom	3+ years	Parent-level pooled credits; capacity priority
GCP	Committed Use Discounts	70%	3 years	Generic vCPU+RAM; suited to variable workloads
GCP	Sustained Use	~30%	None (automatic)	Baseline automatic discount; no commitment

Cloud reservation addresses pricing. It does not address GPU-to-storage throughput, data sovereignty, IP protection, or the fundamental question of whether accelerators are running at capacity. A division can be on a fully optimized Reserved Instance while running GPUs at 30% utilization because the storage layer is the bottleneck. These are independent variables, and both require deliberate action.

A deeper structural issue persists. Cloud reservation mechanisms lock the organization into a provider’s pricing trajectory and capacity roadmap. Reserved Instances require committing to specific instance types in specific regions—a rigidity that conflicts with the evolving geography of pharmaceutical AI workloads. Managed cloud services introduce migration complexity and data-gravity effects that accumulate as hidden debt: the cost of leaving grows with every terabyte ingested. The alternative is an on-premises storage model with consumption-based pricing: capacity consumed at a fixed rate per gigabyte, no egress charges, no idle-resource waste, and no repatriation cost when workloads move between geographies. The economic certainty that cloud reservation approximates, a well-structured on-premises subscription delivers natively.

A global pharma with 200+ cloud accounts, of which 30–40% are unconsolidated, can typically recover CHF 5–15 million annually through contract consolidation alone—before any workload optimization or storage intervention. This is an organizational exercise, not a technical migration.

Part 3 — Five-Country Infrastructure Assessment

A global pharmaceutical enterprise does not choose a single geography. It operates across several simultaneously, each with distinct energy economics, grid constraints, regulatory frameworks, and data sovereignty requirements. This section assesses each from the perspective of an infrastructure executive evaluating where and how to deploy AI workloads—and what the data layer must deliver in each location.

One cross-cutting regulatory development warrants attention before examining individual geographies. The European Commission’s EU GMP Annex 22 on Artificial Intelligence is now in consultation, establishing for the first time that AI tools used in pharmaceutical manufacturing and quality systems must be governed within the pharmaceutical quality system itself—with full lifecycle management, validation, traceability, and auditability. This is not a future consideration; it is an active regulatory workstream that will shape infrastructure requirements in every geography where GxP-regulated operations exist. The infrastructure serving AI in pharmaceutical environments must be designed for compliance from the outset—not retrofitted after deployment.

3.0 The Grid as Binding Constraint

Before assessing individual geographies, one cross-cutting reality requires attention. The global electrical grid is now the binding constraint on AI infrastructure expansion. Transformer lead times have doubled to tripled since 2020. Gartner projects power shortages will restrict 40% of AI data centers by 2027.

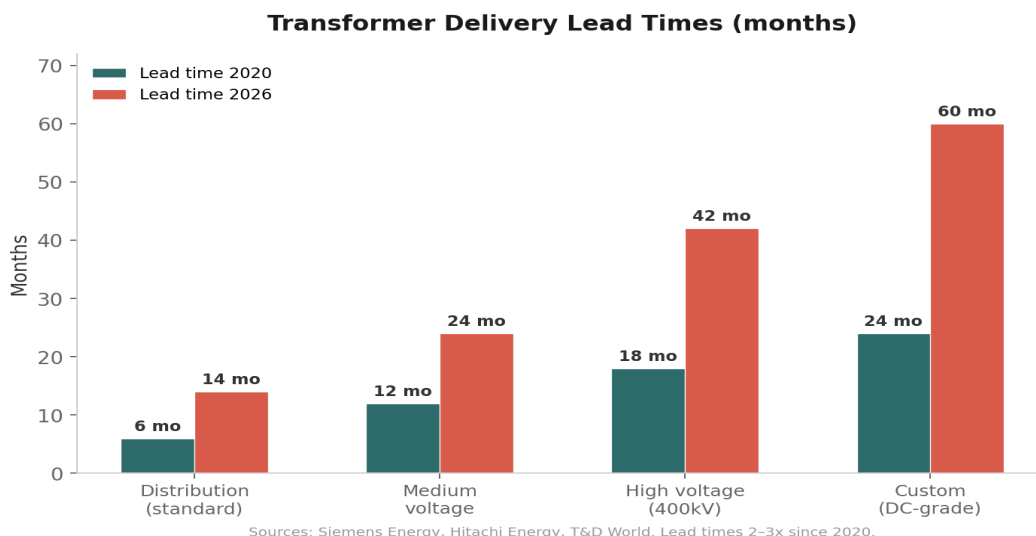


Figure 3 — Transformer delivery lead times (months), 2020 vs. 2026. Sources: Siemens Energy, Hitachi Energy, T&D; World.

In the PJM interconnection (eastern US), data center demand contributed \$9.3 billion to a \$14 billion capacity bill for 2025–2026. IEA projects global data center consumption reaching 945 TWh by 2030, up from 415 TWh in 2024. This grid reality makes on-premises energy efficiency—measured in effective capacity per watt—a strategic differentiator rather than an operational detail.

3.1 United States

Infrastructure

Approximately 5,400 data centers, representing ~40% of global capacity. Grid stress is acute—the PJM region alone saw data centers add \$9.3 billion to capacity costs. Wholesale electricity is projected at \$51/MWh in 2026 (EIA). Data center electricity consumption stands at ~4.4% of the national total, projected to reach 9–12% by 2030.

Pharma Operations

The US hosts the largest concentration of pharmaceutical R&D campuses, including major hubs in the San Francisco Bay Area, the Boston/Cambridge corridor, and the New Jersey pharmaceutical belt. Multiple Top-10 pharma companies have announced multi-billion-dollar US investment programs over the past 24 months. Manufacturing expansion is underway across the Midwest and Southeast.

Regulatory Framework

No federal data protection law equivalent to GDPR. HIPAA governs patient data. The CLOUD Act (2018) enables US authorities to compel disclosure of data held by American companies, including data stored abroad—a material consideration for any pharmaceutical organization hosting proprietary compounds or clinical data on US-headquartered cloud infrastructure.

Data Infrastructure Requirements

Highest data velocity among all five geographies. Genomic sequencing, clinical NLP, drug discovery training, and commercial analytics all generate extreme I/O demands. The storage platform serving US R&D must sustain massively parallel throughput across mixed file sizes, support multi-tenant GPU clusters serving multiple teams simultaneously, and be certified for the NVIDIA DGX SuperPOD architectures that anchor most enterprise AI factories. CLOUD Act exposure must be factored into any decision about where proprietary compound data resides.

3.2 Switzerland

Infrastructure

Approximately 120 data centers with 10+ under construction. Hydro-nuclear energy mix provides low-carbon, stable grid supply. Data centers account for 6–8% of national electricity, with projections suggesting 15% by 2030. Political stability and regulatory predictability make Switzerland one of the highest DC-per-capita countries globally.

Pharma Operations

Basel is the global headquarters for multiple Top-10 pharmaceutical companies and the operational base for several major generics manufacturers. The ETH Zürich / EPFL / Biopôle Lausanne ecosystem drives life sciences innovation. Manufacturing operations cluster around the greater Basel area. Switzerland functions as the strategic decision-making center for global AI programs.

Regulatory Framework

The revised Federal Data Protection Act (nDSG, September 2023) closely aligns with GDPR. Swissmedic provides GxP regulatory alignment. Critically, Switzerland is not an EU member—data cannot be freely transferred to EU data centers without adequacy assessment. Domestically hosted data is not subject to the CLOUD Act. The Federal Council plans new AI regulations by end of 2026.

Data Infrastructure Requirements

The Swiss headquarters campus is where global AI strategy is operationalized: molecular simulation, pharmacovigilance, foundation model development, and enterprise analytics. Compact campus footprints make storage density critical—effective capacity per watt becomes a space constraint, not an efficiency metric. The storage platform must support DGX SuperPOD-certified AI factory configurations alongside mission-critical transactional workloads (clinical trial databases, LIMS). A single platform spanning both workload types reduces vendor complexity at the strategic center of the organization.

3.3 Germany

Infrastructure

Germany hosts approximately 490 data centers—the largest installed base in continental Europe. Frankfurt is the dominant hub but increasingly grid-constrained; NTT is building a 480 MW facility in Nierstein (2026), equivalent to the demand of 500,000 households. The Energy Efficiency Act mandates 100% renewable electricity for data centers by 2027 and PUE ≤ 1.3 by 2030. The government targets an industrial electricity price of ~€50/MWh (subsidized) from 2026.

Pharma Operations

Major pharmaceutical manufacturing and R&D operations across the Rhine-Neckar and upper Rhine regions. Germany is the primary European manufacturing geography for several Top-10 pharma companies. GDPR applies directly, supplemented by the Federal Data Protection Act (BDSG).

Data Infrastructure Requirements

Manufacturing optimization AI—predictive quality, yield improvement, supply chain intelligence—is the dominant workload. These generate mixed I/O patterns: high-frequency sensor data, batch records, and image-based quality inspection. The storage platform must handle heterogeneous I/O natively and support SAP HANA and MES databases at mission-critical availability levels. Energy efficiency directly addresses Germany's strict PUE ≤ 1.3 mandate—the storage layer's power consumption is a compliance variable, not just an operating cost.

3.4 France

Infrastructure

France operates 63 GW of nuclear capacity, supplying approximately 70% of national electricity. Baseload nuclear pricing averages ~€42/MWh—the lowest in Western Europe and the most structurally predictable. The government has announced substantial AI investment commitments. France has established the most advanced sovereign cloud regulatory architecture in Europe: SecNumCloud (highest EU security certification) and HDS (Health Data Hosting, mandatory for personal health data).

Pharma Operations

France is home to several major pharmaceutical headquarters (notably Sanofi) and hosts commercial, medical affairs, and some manufacturing operations for global pharma. It is not, however, a primary R&D or manufacturing geography for Swiss-headquartered pharmaceutical companies, whose operations concentrate in Switzerland, the US, Germany, and increasingly China. France's relevance to the global pharma infrastructure discussion is therefore primarily as an EU data jurisdiction and a potential location for AI training workloads rather than as a core operational site.

Data Infrastructure Requirements

France's nuclear baseload makes it an attractive European location for sustained, energy-intensive AI training workloads where predictable power cost over multi-year horizons is a differentiator. HDS certification is mandatory for any system processing French patient data—a constraint that must be factored into any deployment involving French clinical or commercial data. The SecNumCloud and HDS frameworks align with on-premises and compliance-ready storage architectures.

France is included in this assessment for completeness as a significant EU data jurisdiction. It is not a primary operating geography for the Swiss-headquartered pharmaceutical enterprises that are the principal audience of this report.

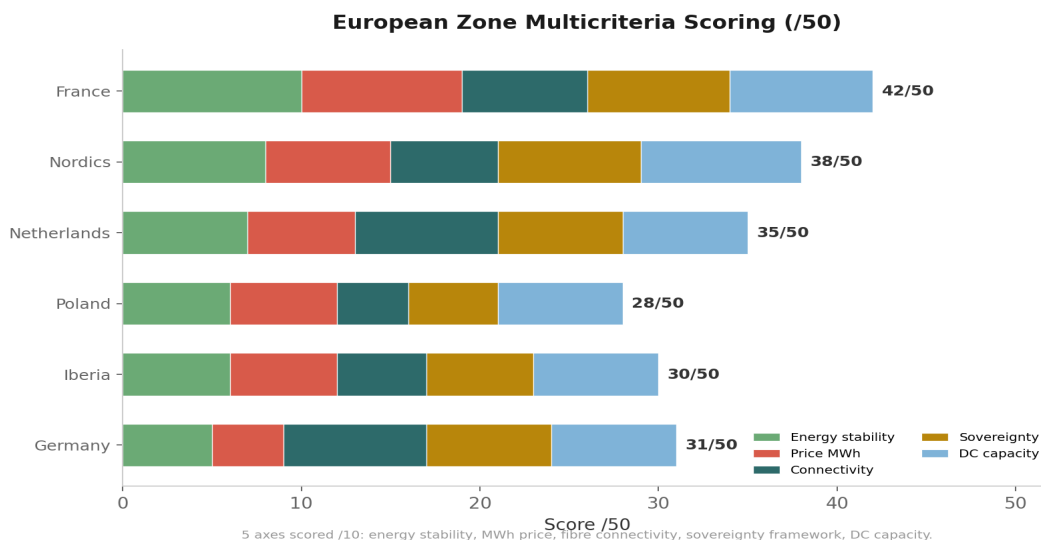


Figure 4 — European zone multicriteria scoring (/50). Axes: energy stability, MWh price, connectivity, sovereignty, DC capacity.

3.5 China

Infrastructure

China mandates strict data localization under the Cybersecurity Law (CSL, amended effective January 2026), the Data Security Law (DSL), and the Personal Information Protection Law (PIPL). Cross-border data transfer requires security assessment by the Cyberspace Administration. The CSL amendments explicitly embed AI governance provisions, including risk assessment, ethics review, and classification requirements for pharmaceutical data deemed “important data.” The AI Plus Action Plan (August 2025) targets 70% AI penetration in key sectors by 2027 and 90% by 2030.

Pharma Operations

Multiple Top-10 pharmaceutical companies operate R&D centres and manufacturing hubs in Shanghai and Beijing. China hosted approximately one-third of global clinical trials in 2024. US export controls restrict sales of high-end NVIDIA GPUs (A100/H100/H200) to China, creating a bifurcated infrastructure requirement where Chinese operations must use approved alternatives.

Data Infrastructure Requirements

Fully localized storage is not optional—it is a compliance requirement. Clinical trial data, patient records, and manufacturing quality data cannot leave Chinese jurisdiction. FlashBlade deployed locally provides the mandated data residency while maintaining performance that keeps GPU-constrained clusters running at maximum utilization. In an environment where GPU supply is limited by export controls, storage throughput becomes even more critical to extracting value from available compute.

3.6 Comparative Summary

Factor	USA	Switzerland	Germany	France	China
Energy cost (industrial)	\$51/MWh	Competitive (hydro)	~€50/MWh (target)	~€42/MWh (nuclear)	Regional (¥0.35–0.65/kWh)
Grid risk	High (PJM saturated)	Moderate (6–8%)	High (Frankfurt)	Low (nuclear baseload)	Moderate (regional)
Data sovereignty	CLOUD Act; HIPAA	nDSG; no CLOUD Act	GDPR + BDSG	GDPR + HDS + SecNumCloud	CSL + DSL + PIPL; mandatory localization

Factor	USA	Switzerland	Germany	France	China
Storage density imperative	Moderate	Critical (compact campus)	High (PUE ≤ 1.3)	Moderate	High (GPU-restricted)
Primary storage requirement	Max-throughput AI/HPC	DGX-certified + transactional	Mixed mfg I/O + SAP HANA	EU sovereign; HDS-compliant	Fully localized; export-compliant

CLOUD Act; HIPAA

nDSG; no CLOUD Act

GDPR + BDSG

GDPR + HDS + SecNumCloud

CSL + DSL + PIPL; mandatory localization

Storage density imperative

Moderate

Critical (compact campus)

High (PUE ≤ 1.3)

Moderate

High (GPU-restricted)

Primary storage requirement

Max-throughput AI/HPC

DGX-certified + transactional

Mixed mfg I/O + SAP HANA

EU sovereign; HDS-compliant

Fully localized; export-compliant

3.7 The Architecture Logic: Cloud, Colocation, On-Premises

Each geography assessed above presents a distinct infrastructure equation. But across all five, the same architectural question recurs—and it is the question that Everpure's Enterprise Data Cloud (EDC) architecture answers: what belongs in the cloud, what belongs in colocation or hosted private cloud, and what belongs on-premises? The answer is not ideological—it is economic, operational, and regulatory.

Cloud: Elasticity at a Price

Public cloud serves a genuine function: burst capacity for unpredictable workloads, rapid prototyping, and geographic reach for globally distributed teams. For a subsidiary launching a new analytics initiative or an R&D team testing a hypothesis, cloud's immediate availability is valuable. But cloud pricing is now structurally inflationary. The extraction-business model described in Part 1 means that sustained workloads on cloud infrastructure operate on the provider's economic terms, not the customer's. Cloud is OpEx by nature, but OpEx that inflates unpredictably is not the same as OpEx that is controlled. For a pharmaceutical enterprise with multi-year AI programs, cloud is appropriate for the variable margin—not for the predictable base.

Colocation and Hosted Private Cloud: The Middle Layer

Colocation facilities offer physical infrastructure—power, cooling, connectivity, physical security—without the customer owning the building. For AI workloads specifically, colocation addresses a challenge that many pharmaceutical campuses were never designed for: the power density and cooling capacity that GPU clusters require. A single DGX SuperPOD rack can draw 40–80 kW—several times the power density of a

conventional server rack. Purpose-built colocation facilities are engineered for these thermal loads from the outset, with redundant utility feeds, high-density liquid cooling infrastructure, and long-term power purchase agreements that provide cost predictability.

For organizations that need to deploy in a specific geography (for sovereignty, latency, or regulatory reasons) but lack campus capacity—or whose campus infrastructure cannot be retrofitted economically for AI-class power density—colocation provides the envelope. Hosted private cloud extends this with managed compute and networking. The customer retains full control of the software, data, and storage stack while the colocation provider absorbs the physical plant risk. In Germany, where grid-constrained Frankfurt makes new on-campus data center builds impractical, and in Switzerland, where compact pharmaceutical campuses face physical expansion limits, colocation may be the most viable path for near-term AI capacity.

On-Premises: The Predictable Core

On-premises infrastructure serves the workloads that define the organization's competitive position: proprietary drug discovery pipelines, clinical data under sovereignty constraints, manufacturing AI tightly coupled to production lines, and enterprise models trained on decades of proprietary data. These workloads are predictable in volume, sustained in duration, and sensitive in content. They are precisely the workloads that extractive cloud economics are designed to monetize most aggressively.

On-premises infrastructure for these core workloads delivers lower, fixed, non-inflationary cost per unit of compute. GPU and CPU acquisition is CapEx—amortized over 3–5 years at a known cost. There is no egress charge, no idle-resource billing, no scarcity premium, no provider debt structure embedded in the price.

The risks are real and should be quantified, not dismissed. First, power and cooling: adapting existing on-campus facilities for AI-class power density can require substantial capital investment—electrical panel upgrades, backup generator capacity, precision cooling systems, and potentially structural reinforcement. Lead times for electrical infrastructure upgrades can exceed 12–18 months, and costs can rival the compute investment itself. If the campus cannot support the thermal load, on-premises deployment may not be feasible without a parallel colocation strategy. Second, technology obsolescence: on-premises hardware ages, requires refresh cycles, and accumulates technical debt if those cycles are deferred. Both risks must be modeled explicitly in the TCO comparison described in Part 5.

These two risks can be managed independently. Power and cooling constraints are addressed through facility assessment, colocation partnerships, or campus upgrades—decisions made at the building level. Technology obsolescence is addressed at the platform level through lifecycle management. An organization can deploy its storage and compute in a colocation facility and still operate it as owned, sovereign infrastructure under its own control—capturing the power and cooling advantages of purpose-built facilities while retaining the economic and operational benefits of on-premises architecture.

Where Storage Changes the Equation

The infrastructure conversation now shifts from compute to data. The technical debt risk of on-premises infrastructure is real—but it is concentrated in the storage layer, not the compute layer. GPUs and CPUs follow Moore's Law variants: they improve on predictable cycles, and replacement is a hardware swap. Storage, by contrast, is entangled with data: migration, revalidation, replication, and integrity verification make storage refresh the most disruptive and costly infrastructure event in pharmaceutical IT.

A storage platform that eliminates this refresh cycle—that is perpetually current, non-disruptively upgraded, and subscription-based—neutralizes the primary objection to on-premises infrastructure. It converts the CapEx liability of aging storage into an OpEx subscription with guaranteed performance, while the compute layer remains on a conventional CapEx refresh cadence that is operationally straightforward.

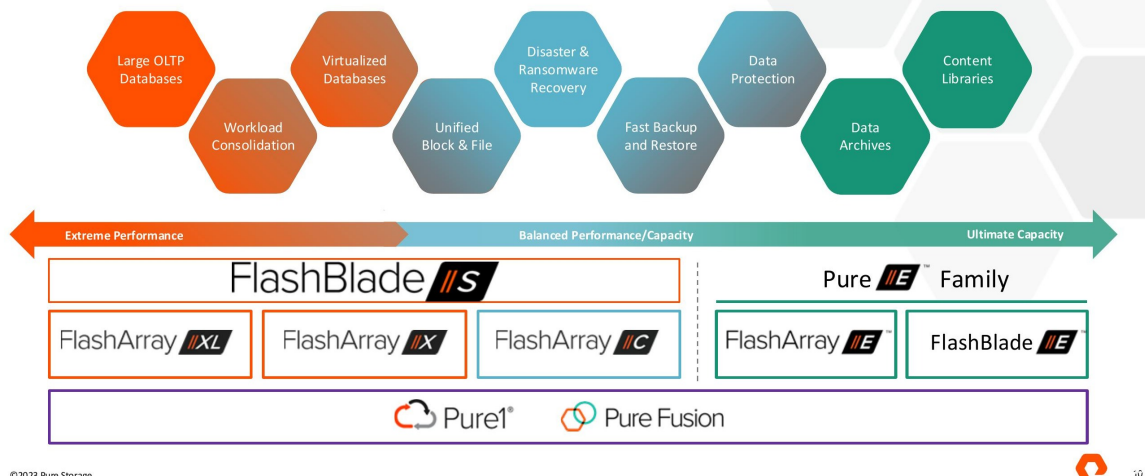
Simultaneously, the right storage platform addresses multiple dimensions that cloud cannot: data sovereignty (the data never leaves the jurisdiction), energy efficiency (all-flash at 1.4 TB/watt reduces per-use-case power cost when GPUs are running at capacity rather than idling on storage waits), SLA guarantees with GPU bandwidth commitments (not subject to cloud throttling during peak demand), and

cyber resilience through immutable snapshots that protect the intellectual property underpinning multi-billion-dollar R&D portfolios.

The architecture for each geography, then, is not cloud-versus-on-prem. It is a deliberate allocation: cloud for the elastic variable, on-premises for the predictable core, and a storage platform that makes on-premises economically and operationally superior for the workloads that matter most.

From Extreme Performance to Ultimate Capacity

All-flash storage ranging from primary application platforms to always-on data repositories



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Everpure Platform — From Extreme Performance to Ultimate Capacity. Source: Everpure.

Part 4 — The Data Layer: Where AI ROI Is Determined

4.1 From Pilot to Production: Beating the Odds

The executive summary cited the failure rates: 42% of AI initiatives missing ROI, 95% of manufacturing AI pilots shelved, only 11% achieving enterprise-wide implementation. These are not technology failures. They are infrastructure failures. The post-mortem pattern is consistent across the industry: models that performed in the lab faltered on the factory floor, in the clinical pipeline, or at commercial scale because the data infrastructure was not designed for AI throughput, cross-functional data flows, or production-grade governance.

The ZS/Harris Poll survey of 115 pharma technology executives (2025) found that 68% identify neglecting data quality and governance as the primary reason AI initiatives fail. Gartner projects that by 2026, 60% of AI projects lacking AI-ready data will be abandoned outright. CIOs interviewed by ZS were direct: data foundations are not optimized for AI consumption, and most systems still lack the architecture and infrastructure needed to support intelligent workflows—including gaps in AI platforms, real-time data pipelines, and GPU/cloud capacity.

Pharmaceutical organizations operate as ecosystems of semi-autonomous business units—R&D, clinical, manufacturing, supply chain, commercial—each with its own systems, data formats, and governance. AI's value comes from connecting across these domains: correlating manufacturing process data with clinical outcomes, linking genomic data with commercial signals, feeding real-world evidence into regulatory submissions. But the infrastructure serving these domains was built for departmental operations, not enterprise-scale AI.

The resulting failure modes are specific and recurring:

- Data engineering consumes the timeline: 50% of pharma AI projects stall during data engineering, not model development. By the time teams wrangle data into usable form, 60% of project timelines have elapsed (Pharmaphorum, 2026).
- Cultural resistance to data sharing: Scientists resist sharing raw data across functions. Teams prefer reports over machine-readable outputs. Cultural barriers compound technical ones.
- Shadow AI and governance gaps: Only 17% of pharma companies have automated controls to prevent sensitive data leakage through AI tools (Kiteworks, 2025). Employees adopt personal AI tools that deliver faster results than formal initiatives—what Qinecsa's chief product officer described as a "shadow AI economy"—creating ungoverned data flows and compliance exposures across organizational boundaries.
- Data readiness: AI models require data in specific, machine-readable formats. Much of pharma's data remains locked in inaccessible formats, legacy systems, or unstructured repositories. Without curation and clinical context, outputs lack relevance.

These challenges are organizational and architectural. No single technology resolves all of them. But the infrastructure layer that connects every other system—where data is stored, governed, moved, and served to compute—is the foundation on which everything else depends. If that layer is fragmented, slow, or designed for a prior era, every intervention built on top of it inherits those limitations.

4.2 The Storage Bottleneck

Within the infrastructure stack, one layer has the most direct and measurable impact on whether AI investments produce returns: storage. The standard enterprise AI deployment proceeds as follows: the organization invests in GPU accelerators—whether NVIDIA DGX, HPE Cray, or Lenovo ThinkSystem clusters—and connects them to existing storage infrastructure. That storage was typically designed for an earlier era: file shares, ERP databases, departmental NAS. It was never architected for the parallel,

sustained, mixed-size I/O patterns that define AI workloads.

The result is endemic and well-documented: GPU utilization rates of 20–40% in environments where storage is the bottleneck. The accelerators are capable; the data path is not. Independent CIO research confirms the pattern: a November 2025 survey of 35 CIOs and CTOs found that while 93% of organizations are actively deploying AI, only 21% report success rates above 80%—with data quality and disconnected systems, not compute or model quality, identified as the primary blockers.

In pharmaceutical workloads, the mismatch is structurally worse:

- Genomics: Hundreds of terabytes per month. File sizes range from kilobyte-scale metadata to multi-gigabyte BAM/CRAM alignments. The storage platform must handle both extremes simultaneously, without tuning or tiering.
- Digital pathology: Whole-slide images of 1–10 GB each, thousands per study. Inference latency is determined entirely by storage read throughput.
- Molecular dynamics and model training: Frequent checkpoint writes during multi-day training runs. Insufficient storage write speed causes training stalls. Lost checkpoints mean lost GPU-hours—unrecoverable.
- Clinical NLP and inference: RAG-based systems querying clinical databases in real time. Sub-millisecond response is required for pharmacovigilance case processing at scale.

Element AI documented this failure mode precisely: GPU utilization of approximately 20% with legacy storage, rising to close to 100% after a purpose-built flash platform was deployed. The storage layer was approximately 10x faster with no specific tuning. A fivefold effective compute gain from a storage intervention alone, with zero additional GPU investment.

ity, performance, cost, or data model reasons, Pure1 centralizes storage
capacity planning for users' entire "fleets" of Pure systems.

pressing data prior to storing it. In addition, FlashArray systems, achieve further
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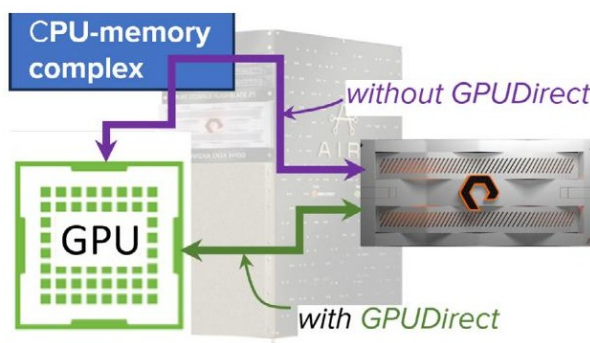


FIGURE 3 Pure-NVIDIA Collaboration

Figure 5 — NVIDIA GPUDirect Storage: Eliminating the CPU Bottleneck. Source: Everpure AI Business White Paper.

4.3 What the Data Platform Must Deliver

The analysis in Sections 4.1 and 4.2 defines a set of requirements that any data platform serving pharmaceutical AI must satisfy. The shift from 'storage platform' to 'data platform' is deliberate: pharmaceutical AI demands not just performance and capacity, but data governance, lifecycle intelligence, and sovereignty controls integrated at the infrastructure layer. These are not product specifications—they are strategic criteria that the executive team should use to evaluate alternatives and hold vendors accountable.

Requirement	Why It Matters for Pharma AI	Assessment Question
Massively parallel I/O	Genomics, pathology, and training workloads generate mixed-size, concurrent I/O that centralised-controller architectures cannot sustain. This is the root cause of GPU idle time.	Does the platform distribute I/O across the full array, or route through controllers that become bottlenecks at scale?
GPU-certified integration	AI factory architectures (DGX SuperPOD, BasePOD) require certified storage to guarantee performance. Uncertified storage introduces latency and support risk.	Is the storage vendor certified by NVIDIA at Foundation, Enterprise, and Cloud Partner levels?
Non-disruptive lifecycle	GxP environments require validation of every infrastructure change. A 3–5 year forklift cycle means revalidation, downtime risk, and technical debt.	Can the platform upgrade controllers, software, and media non-disruptively—without data migration or revalidation?
Consumption-based economics	On-premises infrastructure must deliver cost certainty. Overprovisioning wastes capital; underprovisioning risks performance. Fixed \$/GiB with buffer capacity is the target model.	Does the vendor offer true pay-per-use with GPU bandwidth SLAs and no overprovisioning penalties?
Cyber resilience	Pharma IP (compound structures, trial data, process parameters) is a high-value target. Immutable snapshots that survive administrator compromise are a minimum.	Does the platform provide immutable, indelible snapshots that cannot be deleted by any actor—including compromised credentials?
Energy efficiency	Grid constraints and ESG reporting make per-watt capacity a strategic metric. All-flash at scale consumes 2–5x less power than legacy systems.	What is the platform's effective TB/watt? Does it reduce per-use-case power cost by maximising GPU utilization?
Cross-geography consistency	Five jurisdictions with five regulatory frameworks require a single operational model—same SLA, same management plane, same upgrade path.	Can the vendor deliver a single subscription and SLA framework across every operating geography?
Compute-vendor neutrality	Global pharma deploys different compute platforms per geography. The storage layer must integrate with any accelerator architecture.	Is the storage platform certified across NVIDIA DGX, HPE, and Lenovo—or tied to one compute vendor?
Data intelligence and classification	AI models trained on unclassified, ungoverned data produce unreliable outputs and regulatory exposure. The platform must discover, classify, and contextualize data at the infrastructure layer—mapping to GDPR, PIPL, and GxP frameworks automatically.	Does the platform provide automated, continuous data discovery and classification across all environments—with regulatory mapping integrated at the storage layer rather than bolted on?

AI models trained on unclassified, ungoverned data produce unreliable outputs and create regulatory exposure. Pharmaceutical enterprises must know what data they hold, where it resides, how it is classified under GDPR, PIPL, and GxP, and whether it is ready for AI training or inference—before any model touches it. The platform must discover, classify, and contextualize data at the infrastructure layer, not as a separate toolchain.

4.4 Everpure: The Platform That Meets These Requirements

Everpure (NYSE: PSTG) was founded on a single premise: organizations should manage their data, not their storage. In February 2026, the company formalized that evolution by rebranding from Pure Storage to Everpure—a name that captures its expansion from storage infrastructure into intelligent data management.

The premise has matured into the Enterprise Data Cloud (EDC)—a unified, all-flash platform that consolidates block, file, and object storage across on-premises, hybrid, and cloud environments under a single control plane (Everpure Fusion), automated by policy, orchestrated by software, and accessible via API. Storage becomes invisible; data becomes governable. For pharmaceutical enterprises operating across the regulated geographies described in Part 3, EDC provides the architectural foundation for consistent data sovereignty, security, and performance—regardless of where workloads run.

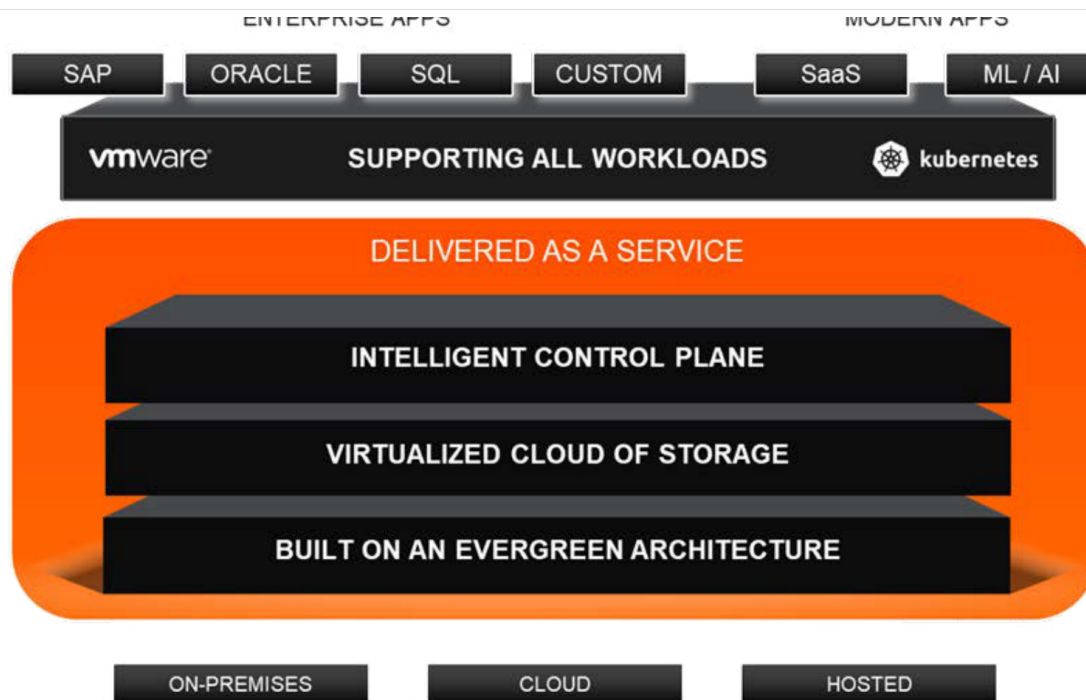


FIGURE 2 The Enterprise Data Cloud architecture.

Figure 7 — The Enterprise Data Cloud Architecture: Evergreen foundation, virtualized storage cloud, intelligent control plane, delivered as a service across on-premises, cloud, and hosted environments. Source: Everpure EDC Architecture White Paper.

For pharmaceutical enterprises, this translates directly to the challenges described in Sections 4.1 through 4.3. The silos between R&D, clinical, manufacturing, and commercial exist in part because the storage infrastructure serving each domain was procured independently, runs on different platforms, and cannot be governed consistently. Everpure resolves this structurally: a unified data layer that spans every workload and every geography, managed as a single estate regardless of where the data physically resides. Named Leader in the 2025 Gartner Magic Quadrant for Enterprise Storage Platforms—positioned highest in execution and furthest in vision—and Leader in Infrastructure Platform Consumption Services. Broadest NVIDIA DGX SuperPOD certification of any storage vendor.

Pharma Application	Workload Type	Platform	Key Differentiator
Drug discovery / LLM training	Sustained high-throughput, multi-day runs, frequent checkpoints	FlashBlade//EXA	Highest-performance AI/HPC; massively parallel data path; eliminates checkpoint stalls
Clinical AI / inference	RAG-based NLP, pharmacovigilance, real-time trial DB queries	FlashBlade//S R2	DGX SuperPOD certified; 16x 400GbE RoCE; sub-ms latency
Molecular sim / digital pathology	Large-file I/O (1–10 GB), GPU-intensive, storage-bound read	FlashBlade//S R2	I/O distributed across full array; no controller bottleneck; 1.4 TB/watt

Pharma Application	Workload Type	Platform	Key Differentiator
Manufacturing AI / ERP / LIMS	Transactional: SAP HANA, MES, batch records, sensor data	FlashArray//X	Six-nines availability; GxP-validated; non-disruptive Evergreen upgrades
Genomic / regulatory archive	Petabyte-scale long-term retention	FlashBlade//E	Cost-optimized archival; WORM-capable; policy-driven tiering
Unified global operations	Cross-geography, cross-workload, single SLA	Evergreen//One	Fixed \$/GiB; GPU bandwidth SLAs; SafeMode cyber resilience; no forklift

Purity OS communicates directly with each flash chip, bypassing legacy SSD firmware. I/O is distributed across the full array rather than routed through centralised controllers—eliminating the metadata bottleneck that degrades performance at AI scale. 16x 400GbE ports per chassis using RDMA over Converged Ethernet (RoCE) on standard enterprise Ethernet, integrating into existing pharma data center networking without rearchitecting. Whether the compute layer is NVIDIA DGX, HPE ProLiant, or Lenovo ThinkSystem, the storage fabric is consistent.

Evergreen//Forever delivers non-disruptive upgrades throughout the platform lifecycle—eliminating 3–5 year forklift cycles and the GxP revalidation burden they carry. Evergreen//One provides the consumption model: fixed rate per gigabyte consumed, buffer capacity included, no overprovisioning penalties, no true-up adjustments. On-premises with Evergreen typically outperforms cloud NPV by 2–3x over five years when GPU utilization gains, energy efficiency, and elimination of egress costs are fully accounted. This is the antithesis of cloud’s hidden migration debt.

Pharmaceutical IP represents billions in future revenue. SafeMode provides immutable, indelible snapshots that cannot be deleted, modified, or encrypted by any actor—including compromised administrator credentials. Combined with rapid-recovery SLAs, this ensures on-premises AI infrastructure can recover to a known-good state without data loss.

1touch extends this resilience posture upstream—from recovery to prevention. Its Access Intelligence capability continuously maps who has access to sensitive data, how they acquired that access, and whether it remains justified. For pharmaceutical enterprises where a single compromised credential can expose proprietary compound structures, clinical trial databases, or manufacturing process parameters, this pre-breach visibility is as critical as post-breach recovery. The combination of 1touch’s access intelligence with SafeMode’s immutable snapshots creates a layered cyber resilience architecture: detect overexposure before it becomes a breach, and recover instantly if one occurs.

Everpure Fusion provides an as-code, API-first management layer that integrates with ServiceNow and existing ITSM workflows. Storage across every geography is provisioned, monitored, and governed from a single operational plane—reducing administrative overhead by an estimated 50–70% versus fragmented multi-vendor environments. This is not a management convenience; it is a prerequisite for the kind of infrastructure governance that pharmaceutical regulatory environments demand.

Cloud-Native Application Layer

For organizations running containerized AI workloads on Kubernetes—increasingly common in drug discovery pipelines, clinical NLP inference, and multi-site trial analytics—Portworx by Everpure extends platform consistency to the application layer. Portworx eliminates the friction between cloud-native development and underlying storage infrastructure: data scientists provision, move, protect, and scale storage through standard Kubernetes interfaces while infrastructure teams retain full governance and policy control. Storage classes, replication policies, and encryption standards follow the workload across on-premises, colocation, and cloud environments—directly supporting the multi-geography architecture described in Part 3. For pharmaceutical enterprises where every AI experiment generates regulated intellectual property, Portworx provides application-aware backup and disaster recovery at the container

level, ensuring data lineage and audit trails survive infrastructure failures without manual intervention.

Contextual Data Intelligence: 1touch

In February 2026, alongside the rebrand, Everpure announced a definitive agreement to acquire 1touch, a data intelligence and orchestration company whose Kontxtual platform provides enterprise-wide data discovery, classification, contextualization, and enrichment. The acquisition closes a critical gap in the Enterprise Data Cloud architecture: the ability to understand data, not just store it. As Everpure's CEO stated, the goal is to help organizations 'not only gain control of their most valuable asset—data—but also understand, enhance, and contextualize that data for actionable intelligence.' The deal is expected to close in Q2 FY2027.

1touch's Kontxtual platform combines AI-driven contextual intelligence, a Multidimensional Data Graph, and policy orchestration to deliver continuous, automated data governance. Independent benchmarks (Tolly Group) validated 98.6% accuracy on structured data discovery across 60 million database rows—scanned in 25 seconds. The platform operates across cloud, SaaS, on-premises, mainframe, and hybrid environments simultaneously, building real-time contextual records that link every sensitive data element to its lineage, usage patterns, access permissions, and risk profile. The technology already powers IBM Guardium Discover and Classify through a global OEM partnership—a credibility marker for pharmaceutical enterprises evaluating vendor maturity.

For pharmaceutical enterprises operating across the five geographies described in Part 3, 1touch addresses the most persistent AI readiness gaps. First, regulatory data classification: Kontxtual automatically maps data elements to GDPR, HIPAA, PIPL, and GxP frameworks, replacing the manual classification audits that consume months and deliver point-in-time snapshots rather than continuous compliance. Second, AI training data provenance: before any model can be trained on clinical, manufacturing, or genomic data, organizations must prove the data's lineage, consent status, and classification—Kontxtual builds this provenance record at the infrastructure layer. Third, data preparation acceleration: the months-long data preparation cycles that account for 60–80% of AI project timelines are driven by the inability to locate, classify, and validate data. Automated discovery and contextual enrichment at the source compresses these cycles from months to days.

Combined with Everpure Fusion's policy-driven governance and the Evergreen architecture's non-disruptive lifecycle, 1touch completes the Enterprise Data Cloud stack: storage, management, classification, sovereignty, and AI readiness—unified under a single platform. The capabilities that were labelled 'future' on Everpure's product roadmap now have a concrete delivery vehicle, converting aspiration into deployable architecture.

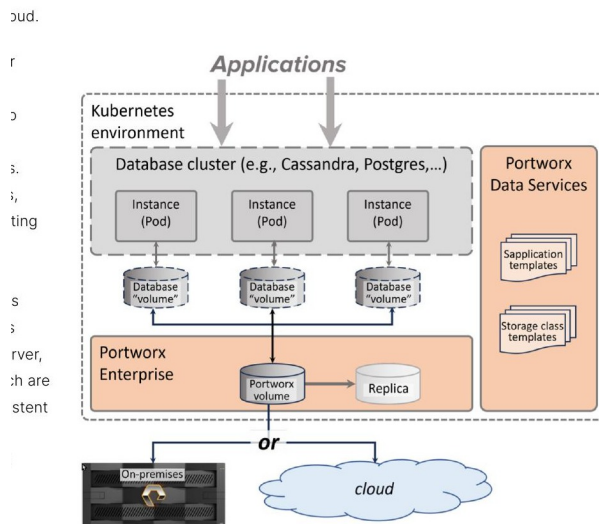


FIGURE 4 Using Portworx with a Database App

concurrently
concurrent jobs that share the same input data is key to rapid implementation. As
simplify deployment of a training application that utilizes data from a database.

allocating its virtual storage devices to either on-premises resources or to a public
ments by copying them to S3 objects in a public or on-premises private cloud.

Figure 6 — Portworx Kubernetes Architecture for AI Workloads. Source: Everpure AI Business White Paper.

Part 5 — From Awareness to Action

5.1 Awareness: Establish the Facts (30 Days)

No strategy survives contact with assumptions. The first step is measurement—not vendor evaluation, not architecture planning, but establishing the facts that will drive internal alignment.

- **GPU utilization audit:** Instrument GPU utilization across every AI cluster globally—headquarters, subsidiaries, acquired entities, regional sites. Below 70% utilization indicates a storage or data-pipeline constraint. This is the single most diagnostic measurement available, and most organizations have never performed it enterprise-wide.
- **Storage I/O profiling:** For every cluster below 70% utilization, capture storage I/O profiles: read/write throughput, latency distribution, metadata operations per second. Determine whether the constraint is storage throughput, data preparation, or network. Everpure offers complimentary infrastructure assessments that benchmark existing storage against FlashBlade I/O profiles for corresponding workload classes—a useful independent data point even at this early stage.
- **Cloud account census:** Identify all subsidiary and divisional cloud accounts operating outside the parent enterprise agreement. Quantify the pricing premium per account. Map which accounts run AI workloads and characterize the storage configuration in each environment.
- **Data sovereignty mapping:** Catalogue where sovereignty-sensitive data resides today: patient records, proprietary compounds, manufacturing process parameters. Identify any instances hosted on CLOUD Act-exposed infrastructure. Map against regulatory requirements per geography. Note that on-premises all-flash storage with immutable snapshots (such as SafeMode) is the only architecture that simultaneously addresses sovereignty, cyber resilience, and regulatory auditability.
- **Data intelligence baseline:** Deploy automated data discovery across a representative subset of storage environments—one geography, two to three workload types. Catalogue what sensitive data exists, where it resides, how it is currently classified, and who has access. This exercise answers the question that precedes all others: do you know what data you have? Organizations that cannot answer this question accurately cannot build a credible AI data readiness strategy. 1touch's Kontxtual platform can execute this baseline in days rather than the months required by manual audits.

These five exercises produce the fact base. They cost nothing except executive attention. They take 30 days. Without them, every subsequent conversation—about architecture, vendors, budgets—is opinion.

5.2 Alignment: Build the Internal Business Case (60–120 Days)

The awareness phase will reveal a gap between current infrastructure performance and what the AI programs require. The alignment phase turns that gap into a business case that CIO, AI leadership, LOB heads, and finance can agree on.

- **TCO and NPV comparison:** Model the total cost of current storage infrastructure plus GPU idle time versus a purpose-built AI storage platform at full GPU utilization. Include energy savings (all-flash at 1.4 TB/watt versus legacy systems), space savings, and critically: the elimination of 3–5 year forklift replacement cycles and their associated GxP revalidation burden. Model per geography. Under a subscription model like Evergreen//One, storage moves off the balance sheet entirely—no depreciating asset, no end-of-life write-down, no capital allocation committee approval for each refresh. The infrastructure is perpetually current as an operating expense. On-premises with this model typically outperforms cloud NPV by 2–3x over a five-year horizon when GPU utilization gains, energy efficiency, and elimination of egress costs are fully accounted.
- **Workload allocation framework:** Based on the geography assessments in Part 3 and the architectural logic in Section 3.7, define which workloads belong on cloud (elastic, variable), colocation

(geography-constrained), and on-premises (predictable core). Align this allocation with budget owners. This is the strategic conversation that prevents fragmented procurement.

- **Storage platform evaluation:** Using the nine criteria from Section 4.3, evaluate current and prospective storage platforms. Score each against the requirements. Everpure is the only platform that satisfies all nine criteria simultaneously—but the framework is designed to be vendor-neutral, and the exercise itself builds internal alignment regardless of the outcome.
- **Regulatory readiness assessment:** Assess the gap between current data governance and what EU GMP Annex 22, PIPL, and evolving FDA AI guidance will require within 24 months. Infrastructure decisions made today must anticipate the regulatory environment of 2028. Non-disruptive lifecycle upgrades (Evergreen//Forever) mean the platform deployed today will meet tomorrow's compliance requirements without revalidation—a material advantage in a regulatory environment that is actively tightening.

5.3 Planning: Design and Pilot (90–150 Days)

With alignment achieved, the organization can design the target architecture and validate it through a controlled pilot.

- **Proof of concept at the point of greatest constraint:** Deploy at the site with the lowest GPU utilization—the highest-ROI intervention point. Validate against an NVIDIA DGX SuperPOD reference architecture. Measure before-and-after GPU utilization. Structure the pilot under Evergreen//One to test the consumption-based OpEx model from day one: fixed rate per gigabyte consumed, buffer capacity included, GPU bandwidth SLAs contractually guaranteed. This pilot simultaneously validates the performance thesis (GPU utilization gain), the economic thesis (OpEx predictability, no CapEx), and the operational thesis (non-disruptive, zero-touch management). It produces the internal business case for global deployment.
- **Cloud contract consolidation:** Migrate subsidiary and divisional cloud accounts into the parent enterprise agreement. Apply existing Reserved Instance and Savings Plan commitments. Target: CHF 5–15 million annual recovery, independent of and additive to any storage optimization.
- **Target architecture definition:** Design the target storage architecture per geography, using the requirements framework from Section 4.3 and the pilot results. The architecture should span the full data temperature spectrum under a single vendor and a single global contract: FlashBlade//EXA for the hottest workloads (drug discovery, LLM training), FlashBlade//S R2 for production AI and inference, FlashArray//X for mission-critical transactional systems, and FlashBlade//E for petabyte-scale genomic and regulatory archives. Evergreen//One wraps all tiers under one subscription, one SLA, and one management plane (Fusion)—eliminating per-geography procurement, per-tier vendor management, and the technical debt that accumulates when different storage platforms serve different workloads.

5.4 Solution Deployment (150–240 Days)

Following pilot validation and architecture design, deploy across remaining priority geographies. Sequence by GPU utilization gap—largest gap represents highest return. Chinese operations require fully localized infrastructure as a compliance obligation under CSL/DSL, not a discretionary decision.

At full scale, the Everpure deployment model delivers three structural advantages that compound over time:

- **Technical debt elimination:** Evergreen//Forever means no hardware ages out, no forklift refresh is ever required, and no GxP revalidation is triggered by infrastructure changes. The technical debt that accumulates in every conventional on-premises environment is structurally eliminated. The platform deployed in year one is the same platform—continuously upgraded—in year five and beyond.
- **Off-balance-sheet asset management:** Under Evergreen//One, all storage across every geography operates as an operating expense. No depreciating assets on the balance sheet. No capital allocation committee approvals for refresh cycles. No end-of-life write-downs. Finance manages a predictable monthly subscription; infrastructure manages capacity and performance. The asset management

burden—procurement, depreciation tracking, disposal, refresh planning—disappears.

- Global contract, global coverage, full data spectrum: A single Evergreen//One contract covers every geography, every workload tier, and every data temperature—from the hottest training clusters (FlashBlade//EXA) through production inference (FlashBlade//S R2) and mission-critical transactional systems (FlashArray//X) to the coldest archival repositories (FlashBlade//E). One vendor, one SLA, one upgrade path, one management plane. For an organization operating across five regulatory jurisdictions, this consistency eliminates the per-geography vendor management, contract negotiation, and operational fragmentation that consume infrastructure leadership bandwidth.

5.5 Self-Assessment: Data Infrastructure Readiness

The following questions can be used by the executive team to assess the organization's current position. Each 'yes' scores one point. Items marked with an asterisk (*) indicate higher-severity exposures.

- Is GPU utilization below 70% in any AI cluster? *
- Is the AI storage layer built on legacy NAS/SAN not designed for parallel AI throughput? *
- Do model training jobs experience checkpoint stalls or I/O wait states?
- Do data science teams report idle time waiting for data during training runs?
- Is the current storage vendor not NVIDIA DGX SuperPOD certified? *
- Are forklift storage replacements planned within 24 months in any geography?
- Does storage infrastructure contribute to unfavourable PUE or ESG metrics?
- Is sovereignty-sensitive data (patient records, proprietary compounds) hosted on CLOUD Act infrastructure? *
- Do subsidiary or acquired-entity AI environments use different storage platforms than headquarters?
- Does the current storage vendor provide contractual GPU bandwidth guarantees? *
- Are Chinese AI workloads running on infrastructure outside Chinese jurisdiction? *

Score 0–4: Managed position. Infrastructure supports current AI programs. Review alignment with growth trajectory and upcoming regulatory requirements.

Score 5–8: Material constraint. GPU investment is underperforming relative to potential. The awareness and alignment phases (5.1–5.2) should begin within 30 days.

Score 9–14: Critical exposure across multiple dimensions. The gap between infrastructure capability and AI program requirements is widening. Immediate executive-level review warranted.

Why Pure?

100% NVMe

Simplicity

Efficiency

Availability 99.9999%

AI&ML Management

Integration

Security

Encryption

Data Reduction

Market Leader

Why Everpure? — 100% NVMe, Simplicity, Efficiency, 99.9999% Availability, AI/ML Management, Integration, Security, Encryption, Data Reduction. Source: Everpure.

Conclusion

This report has examined the pharmaceutical AI infrastructure challenge from four angles: the macro environment reshaping cloud economics, the contract structures that govern enterprise pricing, the geography-specific constraints that determine where and how infrastructure can be deployed, and the data layer that ultimately determines whether AI investments produce returns. Four findings emerge.

The infrastructure gap, not the technology gap, is what separates AI leaders from laggards. 42% of pharma AI initiatives fail to meet ROI expectations. 95% of manufacturing AI pilots are shelved. The consistent root cause is not model quality—it is data readiness—the foundations, governance, and data infrastructure connecting every system. Organizations that solve the data readiness problem first will move from pilot to production while competitors remain in experimentation.

Cloud dependency has become a strategic risk, not just a cost issue. Hyperscalers have become extraction businesses: \$610 billion in 2026 CapEx, negative FCF trajectories, and \$1.5 trillion in projected debt. The pricing implications for enterprise customers with sustained, predictable AI workloads are structural and long-term. An enterprise-wide strategy that deliberately allocates workloads across cloud, colocation, and on-premises—rather than defaulting to cloud—is no longer optional.

The data layer is the highest-leverage intervention point in the infrastructure stack. GPU utilization of 20–40% in storage-constrained environments represents the single largest source of wasted AI investment. Fixing the data layer can deliver a fivefold effective compute gain with zero additional GPU spend. No other infrastructure intervention offers comparable ROI. A data platform that also eliminates lifecycle technical debt, provides cyber resilience, and operates under a single SLA across geographies multiplies that advantage.

Alignment precedes architecture. The most common failure mode is not choosing the wrong technology—it is making infrastructure decisions without cross-functional alignment between CIO, AI leadership, LOB heads, and finance. The awareness → alignment → planning → solution framework in Part 5 exists because the organizational challenge is at least as difficult as the technical one.

Everpure meets every requirement defined in Section 4.3—not as a storage vendor, but as a data platform. Its Enterprise Data Cloud architecture unifies storage, governance, and policy across every deployment environment. With 1touch, it goes further: the platform now discovers, classifies, and contextualizes the data it stores, closing the gap between infrastructure and intelligence that has stalled pharmaceutical AI programs worldwide. No other vendor delivers storage performance, data governance, regulatory classification, cyber resilience, and AI readiness as a single, integrated architecture.

For the executive team accountable for turning pharmaceutical AI investment into competitive advantage, the decision is ultimately about whether the organization's infrastructure will compound in value—or compound in cost. The facts in this report provide the foundation for that decision. The next step is awareness: measure, map, and establish the internal fact base that makes alignment possible.

Glossary

Everpure (PSTG):

Leader, 2025 Gartner Magic Quadrant for Enterprise Storage Platforms. All-flash, purpose-built for AI.

FlashBlade//EXA:

Highest-performance variant engineered for the most demanding AI/HPC workloads.

FlashBlade//S R2:

NVIDIA DGX SuperPOD certified. 16x 400GbE RoCE. Production AI factory storage tier.

FlashBlade//E:

High-density archival storage. Cost-optimized for petabyte-scale long-term retention.

FlashArray//X:

Mission-critical transactional storage for ERP, LIMS, SAP HANA, and clinical trial databases.

Evergreen//One:

Storage-as-a-service subscription. 99.9999% uptime SLA. GPU bandwidth guarantees. No forklift upgrades.

Evergreen//Forever:

Non-disruptive lifecycle upgrade subscription. Always-current infrastructure.

SafeMode:

Immutable snapshot protection. Prevents deletion, modification, or encryption of data—including by compromised administrator credentials. Recovery SLAs included.

Everpure Fusion:

API-first, as-code storage management layer. Unified provisioning and governance across geographies and deployment models. ServiceNow integration.

Purity OS:

Everpure operating system providing massively parallel data path to flash.

DirectFlash:

Proprietary flash module technology bypassing legacy SSD firmware for direct NAND communication.

RoCE:

RDMA over Converged Ethernet. High-performance, low-latency transport over standard Ethernet infrastructure.

nDSG:

Swiss Federal Data Protection Act (revised September 2023). Closely aligned with GDPR.

CSL:

China Cybersecurity Law. Amended effective January 1, 2026. Mandates data localization for “important data.”

PIPL:

China Personal Information Protection Law. China’s equivalent of GDPR.

DSL:

China Data Security Law. Establishes classification and cross-border transfer requirements.

HDS:

French Health Data Hosting certification. Mandatory for systems processing French patient data.

SecNumCloud:

Highest cloud security certification in Europe (issued by French ANSSI).

CLOUD Act:

US Clarifying Lawful Overseas Use of Data Act (2018). Enables US authority access to data held by American companies, including abroad.

GxP:

Good Practice regulations governing pharmaceutical operations (GLP, GCP, GMP, GDP).

DGX SuperPOD:

NVIDIA reference architecture for large-scale enterprise AI clusters.

PUE:

Power Usage Effectiveness. Ratio of total facility power to IT equipment power. 1.0 = theoretical perfect efficiency.

FLAP-D:

Frankfurt, London, Amsterdam, Paris, Dublin. Europe’s traditional data center hub markets.